Core Python Programming Foundation

AI Summer Camp – Day 1



What is Artificial Intelligence (AI) ?



What is AI?

It's the field of creating machines that can **reason, learn, and act** intelligently, often mimicking human cognitive abilities. Think of AI in action:

- **Voice Assistants**
- **Recommendation Systems**
- **Facial Recognition**
- **Self-driving cars**



How do we program it?

AI systems are fundamentally built from the ground up using **code**. To truly understand, build, and innovate in AI, you need to speak its language. We use programming languages like **Python** to:

- **Instruct Machines**
- **Process Data**
- **Design Algorithms**
- **Automate Complex Tasks**

Input Data → Code (Brain) → Intelligent Output

AI is no longer science fiction; it's seamlessly integrated into our daily lives!



python™

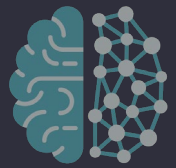
What is python?

Python is a high-level, interpreted, general-purpose programming language

Programming languages translate human logic into machine instructions, with different approaches and complexities. Popular examples include:

- **C++**: System software, game development, embedded systems
- **Java**: Enterprise applications, Android apps
- **JavaScript**: Web development (front-end and back-end)
- **Swift**: iOS mobile applications
- **Python**: AI/Data Science, automation, web development

Human Logic



Machine Instructions

Python is a high-level, interpreted, general-purpose programming language

Python is exceptionally beginner-friendly and powerful, allowing developers to focus on solving problems rather than managing technical complexity. Its versatility allows it to be used across a vast array of applications:

Web Development - Scientific Computing - **AI/Machine Learning** - Automation & Scripting

What makes it general purpose?



1. **Rich Ecosystem:** The Python Package Index (*PyPI*) hosts thousands of pre-built modules that reduce the need to write code from scratch.

1. **Strong Community Support:** Python boasts one of the largest and most active global developer communities, with extensive documentation, tutorials, and constant library growth.

2. **Platform Independence:** Python runs smoothly on various operating systems (Windows, macOS, Linux) with little to no code changes.

Python is a high-level, **interpreted**, general-purpose programming language

Compilation Vs Interpretation

Source code is translated fully into machine code before running *(e.g., C, C++)*	Source code is executed line-by-line at runtime by an interpreter *(e.g., Python, JavaScript)*

Why Python Being Interpreted Is an Advantage

1. **Faster Development**: No compilation step, just write and run
2. **Interactive Development**: Allows for live experimentation
3. **Dynamic Features**: Modify code, types, and objects at runtime
4. **Easier Debugging**: Clear error messages with real-time fixes
5. **Rapid Prototyping**: Ideal for quick tests, data work, and iteration

Python is a **high-level**, interpreted, general-purpose programming language

```cpp
#include <iostream>
#include <vector>
#include <numeric>
#include <iomanip>

int main() {

std::vector<int> numbers = {10, 20, 30, 40, 50};

if (numbers.empty()) {
std::cerr << "Error: Cannot calculate average of empty
vector" << std::endl;
return 1;
}

int sum = std::accumulate(numbers.begin(),
numbers.end(), 0);
double average = static_cast<double>(sum) /
numbers.size();

std::cout << "The average is: " << std::fixed <<
std::setprecision(1) << average << std::endl; return 0;
}
// Output: The average is: 30.0
```

Python abstracts away hardware complexity and memory management, allowing you to write code using human-like syntax and mathematical notation instead of low-level machine instructions.

← **Other Programming Languages (*C++*)** Vs **Python** ↓

```python
numbers = [10, 20, 30, 40, 50]

if not numbers:
    print("Error: Cannot calculate average of
    empty list")

else:
    average = sum(numbers) / len(numbers)

print(f"The average is: {average:.1f}")
# Output: The average is: 30.0
```

Tools used when coding with python

AI Workbench:

A programming environment is the setup where you **write**, **run**, and **test** code. However, setting one up locally can be tricky for beginners

Google Colab is a *cloud-based* environment that needs no setup

Based on **Jupyter** Notebooks, which support:

Code Cells: Run Python code
Text Cells: Add notes using Markdown

Why use google colab?



No Setup Needed

Pre-installed Python + AI libraries



Interactive Format

Mix explanations with live code



Free Hardware Access

Use Google’s powerful GPUs & TPUs



Easy Collaboration

Share and collaborate in real-time

How to get started? colab.research.google.com

